Lecture plan · for the lecturer

# Causal inference: DAGs and difference-in-differences

Data science · University · Lecture 3

| Time | Materials |
| --- | --- |
| 60 min | Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab |

## Learning goals

- Spot confounding.
- Choose an adjustment set from a DAG.
- Estimate an effect with DiD.

## Activity: Draw it, then estimate it

1. Warm-up (5 min): Show the tutoring-centre finding and ask students to vote, then ask who chooses to go.
2. Teach (15 min): Draw the tutoring DAG and mark confounders, mediators and colliders.
3. Draw (15 min): Teams draw a DAG for a headline claim and choose an adjustment set.
4. Estimate (20 min): Compute a DiD from a two-by-two table, then plot pre-trends and discuss whether they convince.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

## Discussion

- What would make you trust the tutoring estimate?
- Why is a collider dangerous to adjust for?
- How will you grade assumptions?

## Exit ticket

- Draw a three-node DAG and say what you would adjust for.

**Try it in your next lecture**
Give teams one news headline that claims a cause; ask them to draw the causal diagram and name one confounder the article ignored.

Seminar sheet · for students

# Causal inference: DAGs and difference-in-differences

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Name _____ Group _____ Date _____

**1** **Treated: 50 to 58. Comparison: 52 to 55. What is the DiD estimate?**

◯ 5 points ◯ 3 points ◯ 8 points

**2** **Work out the difference-in-differences.**

Treated: 40 → 49. Comparison: 45 → 48. The treated change is _____, the comparison change is _____, so the effect is _____.

**Word bank:** 12 · 6 · 3 · 9 · 4

**3** **To estimate the effect of tutoring on grades, which should you adjust for? Choose all that apply.**

☐ Year of study, which affects both ☐ Being on the dean's list, caused by both tutoring and grades

☐ Hours studied, which tutoring changes ☐ Prior grade, which affects both tutoring and grades

**4** **Parallel trends can be confirmed by checking that the two groups had similar trends before the treatment.**

◯ True ◯ False

**5** **Match each design to the threat it most directly addresses.**

| | | |
|---|---|---|
| Randomised experiment | • • • • | Fixed group differences and common shocks |
| Adjusting for measured confounders | • • • • | Hidden, unstated assumptions |
| Difference-in-differences | • • • • | All confounding, measured or not |
| Drawing the DAG first | • • • • | Back-door paths through observed variables |

Answers · for the lecturer

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**1** **5 points**

(58 − 50) − (55 − 52) = 8 − 3 = 5.

**2** **Treated: 40 → 49. Comparison: 45 → 48. The treated change is 9, the comparison change is 3, so the effect is 6.**

49 − 40 = 9, 48 − 45 = 3, and 9 − 3 = 6 points, assuming parallel trends.

**3** **Prior grade, which affects both tutoring and grades; Year of study, which affects both**

Prior grade and year are confounders; hours studied is a mediator and the dean's list is a collider, so adjusting for either biases the estimate.

**4** **False**

Similar pre-trends make the assumption more plausible, but it concerns what would have happened afterwards, which we never observe.

**5** **Randomised experiment — All confounding, measured or not**
**Adjusting for measured confounders — Back-door paths through observed variables**
**Difference-in-differences — Fixed group differences and common shocks**
**Drawing the DAG first — Hidden, unstated assumptions**

Each design buys protection against a different threat; students should name which one their design leaves open.

## For you to think about

**Which causal claim do your students meet most often in your field? What diagram would you ask them to draw for it?**